

ECON 101H, Introduction to Economics

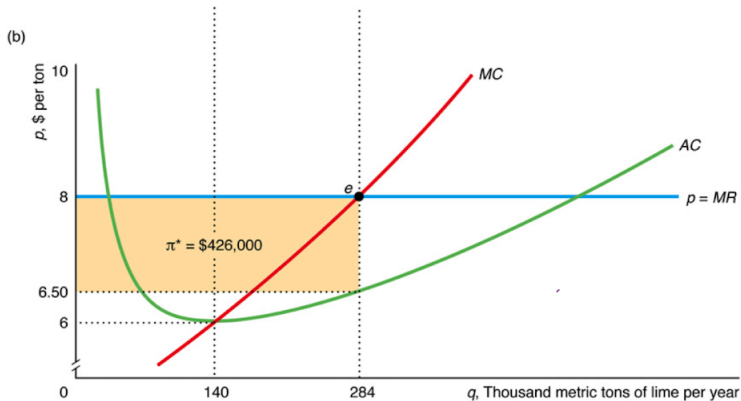
Recitation 5: Long Run Perfect Equilibrium

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- Profit is the rectangle $q^*(p - AC(q^*))$



Short-Run vs. Long-Run Dynamics

A perfectly competitive firm is maximizing its profit in the short-run. At the profit-maximizing level of output, $P < ATC$.

Compared to the short-run, in the long-run there will be _____ firms, and each firm will produce _____ quantity.

- a. more; a smaller
- b. more; a larger
- c. fewer; a smaller
- d. fewer; a larger
- e. fewer; the same

Market Equilibrium Analysis

There are 80 firms in a perfectly competitive market, each with the following total cost function:

$$TC(q) = 144 + 4q^2$$

Market demand is represented by:

$$Q_D = 1600 - 30P$$

1. How much output will each firm produce in the short-run, and how much profit will it make?
2. State what you expect will happen in the long-run to the number of firms, market price, individual quantity, and profit.
3. Calculate each of the above at long-run equilibrium. Calculate CS and PS